

ASSESSMENT – 5(ABHAY KUMAR 25BCE2537)

EXECUTION OF TCL QUERIES

(SET -3-

SALES INVENTORY DATASET)

```
mysql> SET GLOBAL local_infile = 1;
Query OK, 0 rows affected (0.00 sec)

mysql> ^C
mysql> SHOW VARIABLES LIKE 'local_infile';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| local_infile  | ON    |
+-----+-----+
1 row in set (0.01 sec)

mysql> USE retail_tcl;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_retail_tcl |
+-----+
| inventory             |
| inventory_data        |
| orders               |
| retail_store_inventory_data |
+-----+
4 rows in set (0.00 sec)

mysql> LOAD DATA LOCAL INFILE 'C:/Users/Asus/Downloads/retail_store_in
y_data.csv'
-> INTO TABLE inventory_data
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\r\n'
-> IGNORE 1 ROWS
-> (date, store_id, product_id, category, region,
-> inventory_level, units_sold, units_ordered,
-> demand_forecast, price, discount, weather_condition,
-> holiday_promotion, competitor_pricing, seasonality);
Query OK, 73100 rows affected (1.31 sec)
Records: 73100 Deleted: 0 Skipped: 0 Warnings: 0
```

```
mysql> CREATE TABLE inventory (
-> product_id VARCHAR(20) PRIMARY KEY,
-> stock_quantity INT
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO inventory (product_id, stock_quantity)
-> SELECT
-> product_id,
-> MAX(inventory_level)
-> FROM inventory_data
-> GROUP BY product_id;
Query OK, 20 rows affected (0.30 sec)
Records: 20 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM inventory;
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 500            |
| P0002      | 500            |
| P0003      | 500            |
| P0004      | 500            |
| P0005      | 500            |
| P0006      | 500            |
| P0007      | 500            |
| P0008      | 500            |
| P0009      | 500            |
| P0010      | 500            |
| P0011      | 500            |
| P0012      | 500            |
| P0013      | 500            |
| P0014      | 500            |
| P0015      | 500            |
| P0016      | 500            |
| P0017      | 500            |
| P0018      | 500            |
| P0019      | 500            |
| P0020      | 500            |
+-----+-----+
20 rows in set (0.00 sec)
```

1. COMMIT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
  -> SET stock_quantity = stock_quantity - 5
  -> WHERE product_id = 'P0001';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 495           |
+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 495           |
+-----+-----+
1 row in set (0.00 sec)
```

Why?

- COMMIT permanently saves all successful changes in the transaction.

2. ROLLBACK

```
mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 495           |
+-----+-----+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
  -> SET stock_quantity = stock_quantity - 100
  -> WHERE product_id = 'P0001';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 395           |
+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 495           |
+-----+-----+
1 row in set (0.00 sec)
```

Why?

- ROLLBACK cancels all changes made since START TRANSACTION.

- The stock returns to its previous value.

3. SAVEPOINT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
  -> SET stock_quantity = stock_quantity - 1
  -> WHERE product_id = 'P0001';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SAVEPOINT stock_updated;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO payments
  -> (order_id, amount, status)
  -> VALUES
  -> (5001, 1500, 'FAILED');
Query OK, 1 row affected (0.01 sec)

mysql> ROLLBACK TO SAVEPOINT stock_updated;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001      | 485           |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
  -> FROM payments
  -> WHERE order_id = 5001;
Empty set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

The savepoint protects the earlier stock update.

The failed payment is undone, but the stock update remains.

4. — ROLLBACK TO SAVEPOINT

```
mysql> SELECT *
  -> FROM customers
  -> WHERE customer_id = 2001;
+-----+-----+-----+
| customer_id | customer_name | loyalty_points |
+-----+-----+-----+
| 2001        | Customer 2001 | 100           |
+-----+-----+-----+
1 row in set (0.01 sec)

mysql> INSERT INTO orders
  -> (customer_id, order_date, total_amount)
  -> VALUES
  -> (2001, CURDATE(), 500);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT *
  -> FROM orders
  -> ORDER BY order_id DESC
  -> LIMIT 1;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
| 5011     | 2001        | 2026-08-27 | 500.00       | PLACED |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE customers
  -> SET loyalty_points = loyalty_points + 50
  -> WHERE customer_id = 2001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT *
  -> FROM customers
  -> WHERE customer_id = 2001;
+-----+-----+-----+
| customer_id | customer_name | loyalty_points |
+-----+-----+-----+
| 2001        | Customer 2001 | 150           |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SAVEPOINT points_added;
Query OK, 0 rows affected (0.00 sec)
```

Why?

Only the work after points added is undone.

The loyalty-point update remains.

```
mysql> DELETE FROM orders
-> WHERE order_id = 1;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT *
-> FROM orders
-> WHERE order_id = 1;
Empty set (0.00 sec)

mysql> ROLLBACK TO SAVEPOINT points_added;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
-> FROM customers
-> WHERE customer_id = 2001;
+-----+-----+-----+
| customer_id | customer_name | loyalty_points |
+-----+-----+-----+
|          2001 | Customer 2001 |             150 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM orders
-> WHERE order_id = 1;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
|         1 |          2001 | 2025-01-15 |        1500.00 | PLACED |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

5.RELEASE SAVEPOINT

```
mysql> SELECT *
-> FROM products
-> WHERE category = 'Electronics';
+-----+-----+-----+-----+
| product_id | product_name | category | price |
+-----+-----+-----+-----+
|          101 | Laptop | Electronics | 1000.00 |
|          102 | Phone | Electronics | 800.00 |
+-----+-----+-----+-----+
2 rows in set (0.01 sec)

mysql> SELECT
-> product_id,
-> category,
-> price
-> FROM products
-> WHERE category = 'Electronics';
+-----+-----+-----+
| product_id | category | price |
+-----+-----+-----+
|          101 | Electronics | 1000.00 |
|          102 | Electronics | 800.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE products
-> SET price = price * 0.9
-> WHERE category = 'Electronics';
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> SELECT
-> product_id,
-> category,
-> price
-> FROM products
-> WHERE category = 'Electronics';
+-----+-----+-----+
| product_id | category | price |
+-----+-----+-----+
|          101 | Electronics | 900.00 |
|          102 | Electronics | 720.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SAVEPOINT discount_applied;
Query OK, 0 rows affected (0.00 sec)
```

RELEASE SAVEPOINT removes the savepoint because that stage has been successfully completed.

```
mysql> SAVEPOINT discount_applied;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT
  -> product_id,
  -> category,
  -> price
  -> FROM products
  -> WHERE category = 'Electronics';
+-----+-----+-----+
| product_id | category | price |
+-----+-----+-----+
| 101 | Electronics | 900.00 |
| 102 | Electronics | 720.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

6.SET TRANSACTION

```
mysql> SET TRANSACTION ISOLATION LEVEL READ COMMITTED;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0001';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0001 | 485 |
+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

Why?

READ COMMITTED means a transaction can read data committed by other transactions, while uncommitted changes from another transaction are not visible.

7. ORDER PLACEMENT

```
mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0002';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0002 | 500 |
+-----+-----+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
  -> SET stock_quantity = stock_quantity - 3
  -> WHERE product_id = 'P0002';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0002';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0002 | 497 |
+-----+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO orders
  -> (customer_id, order_date, total_amount)
  -> VALUES
  -> (2001, '2025-02-01', 899);
Query OK, 1 row affected (0.00 sec)

mysql> SET @new_order_id = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @new_order_id;
+-----+
| @new_order_id |
+-----+
| 5012 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO payments
-> (order_id, amount, status)
-> VALUES
-> (@new_order_id, 899, 'SUCCESS');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT *
-> FROM payments
-> WHERE order_id = @new_order_id;
+-----+-----+-----+-----+
| payment_id | order_id | amount | status |
+-----+-----+-----+-----+
| 6 | 5012 | 899.00 | SUCCESS |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

Why?

All three operations represent one order. If successful, COMMIT permanently saves them

8. REFUND/CANCELLATION

```
mysql> SELECT *
-> FROM orders
-> ORDER BY order_id DESC
-> LIMIT 1;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
| 5012 | 2001 | 2025-02-01 | 899.00 | PLACED |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM orders
-> WHERE order_id = 2;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
| 2 | 2001 | 2025-01-15 | 1500.00 | PLACED |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM inventory
-> WHERE product_id = 'P0002';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0002 | 497 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
-> SET stock_quantity = stock_quantity + 3
-> WHERE product_id = 'P0002';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE orders
-> SET status = 'CANCELLED'
-> WHERE order_id = 2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SAVEPOINT refund_point;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
-> FROM payments
-> WHERE order_id = 2;
Empty set (0.00 sec)

mysql> ROLLBACK TO SAVEPOINT refund_point;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
-> FROM orders
-> WHERE order_id = 2;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
| 2 | 2001 | 2025-01-15 | 1500.00 | CANCELLED |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM inventory
-> WHERE product_id = 'P0002';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0002 | 503 |
+-----+-----+
1 row in set (0.00 sec)
```

Why?

The stock reversal and cancellation remain, while the failed refund update is undone.

9. BULK PRICE UPDATE WITH SAVEPOINT

```
mysql> SELECT DISTINCT category
-> FROM products;
+-----+
| category |
+-----+
| Electronics |
| Furniture |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT
-> product_id,
-> category,
-> price
-> FROM products
-> WHERE category = 'Furniture';
+-----+-----+-----+
| product_id | category | price |
+-----+-----+-----+
| 103 | Furniture | 200.00 |
| 104 | Furniture | 300.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT before_price_update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE products
-> SET price = price * 1.1
-> WHERE category = 'Furniture';
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0

mysql> SELECT
-> product_id,
-> category,
-> price
-> FROM products
-> WHERE category = 'Furniture';
+-----+-----+-----+
| product_id | category | price |
+-----+-----+-----+
| 103 | Furniture | 220.00 |
| 104 | Furniture | 330.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> ROLLBACK TO SAVEPOINT before_price_update;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT
-> product_id,
-> category,
-> price
-> FROM products
-> WHERE category = 'Furniture';
+-----+-----+-----+
| product_id | category | price |
+-----+-----+-----+
| 103 | Furniture | 200.00 |
| 104 | Furniture | 300.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

Why?

The price update is undone before the transaction commits.

10. MULTI-STEP CHECKOUT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
  -> SET stock_quantity = stock_quantity - 1
  -> WHERE product_id = 'P0003';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0003';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0003      |             499 |
+-----+-----+
1 row in set (0.00 sec)

mysql> SAVEPOINT item1_reserved;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE inventory
  -> SET stock_quantity = stock_quantity - 1
  -> WHERE product_id = 'P0004';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id = 'P0004';
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0004      |             499 |
+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK TO SAVEPOINT item1_reserved;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
  -> FROM inventory
  -> WHERE product_id IN ('P0003', 'P0004');
+-----+-----+
| product_id | stock_quantity |
+-----+-----+
| P0003      |             499 |
| P0004      |             500 |
+-----+-----+
```

Why?

Product P003 remains deducted, while the product P004 deduction is undone.

PRACTICE 1 Customer + Order

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO customers
  -> (customer_id, customer_name, loyalty_points)
  -> VALUES
  -> (3001, 'Practice Customer', 0);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
  -> (customer_id, order_date, total_amount)
  -> VALUES
  -> (3001, CURDATE(), 500);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM customers WHERE customer_id = 3001;
+-----+-----+-----+
| customer_id | customer_name | loyalty_points |
+-----+-----+-----+
|          3001 | Practice Customer |             0 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM orders WHERE customer_id = 3001;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
|       5003 |          3001 | 2026-08-25 |         500.00 | PLACED |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```


2. Update then ROLLBACK

```
mysql> SELECT price
-> FROM products
-> WHERE product_id = 101;
Empty set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE products
-> SET price = price + 100
-> WHERE product_id = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql>
mysql> SELECT price
-> FROM products
-> WHERE product_id = 101;
Empty set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT price
-> FROM products
-> WHERE product_id = 101;
Empty set (0.00 sec)
```

3. Delete order item and COMMIT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM order_items
-> WHERE order_id = 5001;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
-> FROM order_items
-> WHERE order_id = 5001;
Empty set (0.00 sec)

mysql> |
```

4. Payment then complete ROLLBACK

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO payments
-> (order_id, amount, status)
-> VALUES
-> (5001, 1500, 'SUCCESS');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT *
-> FROM payments
-> WHERE order_id = 5001;
+-----+-----+-----+-----+
| payment_id | order_id | amount | status |
+-----+-----+-----+-----+
|          2 |       5001 | 1500.00 | SUCCESS |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
-> FROM payments
-> WHERE order_id = 5001;
Empty set (0.00 sec)
```

5. Order survives, payment disappears

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
  -> (customer_id, order_date, total_amount)
  -> VALUES
  -> (2001, CURDATE(), 750);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT order_created;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO payments
  -> (order_id, amount, status)
  -> VALUES
  -> (LAST_INSERT_ID(), 750, 'FAILED');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT order_created;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

6. Full order placement

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
  -> (customer_id, order_date, total_amount)
  -> VALUES
  -> (2001, CURDATE(), 1200);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SET @new_order_id = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_items
  -> (order_id, product_id, price)
  -> VALUES
  -> (@new_order_id, 101, 1200);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO order_payments
  -> (order_id, payment_value, payment_status)
  -> VALUES
  -> (@new_order_id, 1200, 'SUCCESS');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> |
```

7. Failed payment with partial rollback

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
      -> (customer_id, order_date, total_amount)
      -> VALUES
      -> (2002, CURDATE(), 899);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SET @new_order_id = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_items
      -> (order_id, product_id, price)
      -> VALUES
      -> (@new_order_id, 205, 899);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT before_payment;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_payments
      -> (order_id, payment_value, payment_status)
      -> VALUES
      -> (@new_order_id, 899, 'FAILED');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT before_payment;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> |

```

8. Order cancellation

```

mysql> SELECT *
      -> FROM inventory
      -> WHERE product_id = 205;
+-----+-----+
product_id | stock_quantity |
+-----+-----+
      205 |             20 |
+-----+-----+
row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE inventory
      -> SET stock_quantity = stock_quantity + 3
      -> WHERE product_id = 205;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE orders
      -> SET status = 'CANCELLED',
      ->      order_status = 'CANCELLED'
      -> WHERE order_id = 5010;
ERROR 1054 (42S22): Unknown column 'order_status' in 'field list'
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

9. Partial refund

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE order_payments
      -> SET payment_value = payment_value - 100
      -> WHERE order_id = 5001;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SAVEPOINT refund_applied;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_reviews
      -> (order_id, review_note, status)
      -> VALUES
      -> (5001, 'Partial refund processed', 'UPDATED');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT refund_applied;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

10. Multi-item cart

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_items
      -> (order_id, product_id, price)
      -> VALUES
      -> (5001, 101, 1000);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT item1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_items
      -> (order_id, product_id, price)
      -> VALUES
      -> (5001, 102, 800);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT item2;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_items
      -> (order_id, product_id, price)
      -> VALUES
      -> (5001, 103, 200);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT item2;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

11. Three savepoints

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO customers
  -> (customer_id, customer_name, loyalty_points)
  -> VALUES
  -> (3002, 'Three Savepoint Customer', 0);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT customer_added;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
  -> (customer_id, order_date, total_amount)
  -> VALUES
  -> (3002, CURDATE(), 500);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT order_added;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO payments
  -> (order_id, amount, status)
  -> VALUES
  -> (LAST_INSERT_ID(), 500, 'SUCCESS');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT payment_added;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT order_added;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

```

12. SELLER DISPUTE

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE order_items
  -> SET price = price + 50
  -> WHERE order_id = 5001;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql>
mysql> SAVEPOINT price_changed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sellers
  -> SET seller_rating = seller_rating - 0.5
  -> WHERE seller_id = 1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SAVEPOINT rating_changed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO order_reviews
  -> (order_id, review_note, status)
  -> VALUES
  -> (5001, 'Seller dispute review', 'NEW');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT rating_changed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;

```

13. UPDATE + DELETE with savepoint

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE order_payments
    -> SET payment_value = payment_value + 50
    -> WHERE order_id = 5001;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SAVEPOINT payment_updated;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM order_reviews
    -> WHERE order_id = 5001;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT payment_updated;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> |

```

14. RELEASE SAVEPOINT

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE inventory
    -> SET stock_quantity = stock_quantity - 2
    -> WHERE product_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT stock_confirmed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> RELEASE SAVEPOINT stock_confirmed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
    -> (customer_id, order_date, total_amount)
    -> VALUES
    -> (2001, CURDATE(), 200);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

15. READ COMMITTED

```
mysql> SET TRANSACTION ISOLATION LEVEL READ COMMITTED;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT *
      -> FROM orders;
+-----+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount | status |
+-----+-----+-----+-----+-----+
| 1 | 2001 | 2025-01-15 | 1500.00 | PLACED |
| 2 | 2001 | 2025-01-15 | 1500.00 | PLACED |
| 5001 | 2001 | 2025-01-15 | 1500.00 | PLACED |
| 5002 | 2002 | 2025-02-01 | 899.00 | PLACED |
| 5003 | 3001 | 2026-08-25 | 500.00 | PLACED |
| 5004 | 2001 | 2026-08-25 | 750.00 | PLACED |
| 5005 | 2001 | 2026-08-25 | 1200.00 | PLACED |
| 5006 | 2002 | 2026-08-25 | 899.00 | PLACED |
| 5007 | 3002 | 2026-08-25 | 500.00 | PLACED |
| 5008 | 2001 | 2026-08-25 | 200.00 | PLACED |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

16. Three products, one unavailable

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE inventory
      -> SET stock_quantity = stock_quantity - 1
      -> WHERE product_id = 301;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT item1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE inventory
      -> SET stock_quantity = stock_quantity - 1
      -> WHERE product_id = 302;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT item2;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE inventory
      -> SET stock_quantity = stock_quantity - 1
      -> WHERE product_id = 999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> ROLLBACK TO SAVEPOINT item2;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

17. Bulk price update, full ROLLBACK

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE products
    -> SET price = price * 0.8
    -> WHERE category = 'Electronics';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SELECT *
    -> FROM products
    -> WHERE category = 'Electronics';
Empty set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
    -> FROM products
    -> WHERE category = 'Electronics';
Empty set (0.00 sec)

```

18. Full refund

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE order_payments
    -> SET payment_status = 'REFUNDED'
    -> WHERE order_id = 5010;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> UPDATE inventory
    -> SET stock_quantity = stock_quantity + 3
    -> WHERE product_id = 205;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE orders
    -> SET status = 'REFUNDED',
    ->     order_status = 'REFUNDED'
    -> WHERE order_id = 5010;
ERROR 1054 (42S22): Unknown column 'order_status' in 'field list'
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

```


19. Delete spam review

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SAVEPOINT before_review_delete;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT *
      -> FROM order_reviews
      -> WHERE review_id = 1;
Empty set (0.00 sec)

mysql>
mysql> DELETE FROM order_reviews
      -> WHERE review_id = 1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

20. Flash Sale — Savepoint Strategy for Order #37 Failure

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
      -> (customer_id, order_date, total_amount)
      -> VALUES
      -> (2001, CURDATE(), 100);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT order_36_completed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO orders
      -> (customer_id, order_date, total_amount)
      -> VALUES
      -> (2002, CURDATE(), 200);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT order_36_completed;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```

mysql> USE retail_tcl;
Database changed
mysql>
mysql> CREATE TABLE IF NOT EXISTS inventory (
    ->     product_id INT PRIMARY KEY,
    ->     stock_quantity INT
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS customers (
    ->     customer_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(100),
    ->     loyalty_points INT DEFAULT 0
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS orders (
    ->     order_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     customer_id INT,
    ->     order_date DATE,
    ->     total_amount DECIMAL(10,2),
    ->     status VARCHAR(30) DEFAULT 'PLACED',
    ->     order_status VARCHAR(30) DEFAULT 'PLACED'
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS payments (
    ->     payment_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT,
    ->     amount DECIMAL(10,2),
    ->     status VARCHAR(30)
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS products (
    ->     product_id INT PRIMARY KEY,
    ->     product_name VARCHAR(100),
    ->     category VARCHAR(50),
    ->     price DECIMAL(10,2)

```

```

mysql> CREATE TABLE IF NOT EXISTS payments (
    ->     payment_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT,
    ->     amount DECIMAL(10,2),
    ->     status VARCHAR(30)
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS products (
    ->     product_id INT PRIMARY KEY,
    ->     product_name VARCHAR(100),
    ->     category VARCHAR(50),
    ->     price DECIMAL(10,2)
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS order_items (
    ->     order_item_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT,
    ->     product_id INT,
    ->     price DECIMAL(10,2)
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS order_payments (
    ->     order_payment_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT,
    ->     payment_value DECIMAL(10,2),
    ->     payment_status VARCHAR(30)
    -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE IF NOT EXISTS order_reviews (
    ->     review_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT,
    ->     review_note VARCHAR(255),
    ->     status VARCHAR(30)
    -> );

```

ASSESSMENT -5(25BCE2537 ABHAY KUMAR)

DATABASE SYSTEMS(SET – 3 WORLD POPULATION DATASET)

- ARITHMETIC OPERATION

```
mysql> CREATE TABLE world_population (
  ->   `Rank` INT,
  ->   `CCA3` VARCHAR(10),
  ->   `Country/Territory` VARCHAR(100),
  ->   `Capital` VARCHAR(100),
  ->   `Continent` VARCHAR(50),
  ->   `2022 Population` BIGINT,
  ->   `2020 Population` BIGINT,
  ->   `2015 Population` BIGINT,
  ->   `2010 Population` BIGINT,
  ->   `2000 Population` BIGINT,
  ->   `1990 Population` BIGINT,
  ->   `1980 Population` BIGINT,
  ->   `1970 Population` BIGINT,
  ->   `Area (km²)` BIGINT,
  ->   `Density (per km²)` DECIMAL(15,4),
  ->   `Growth Rate` DECIMAL(10,6),
  ->   `World Population Percentage` DECIMAL(10,4)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_worlddb |
+-----+
| world_population |
+-----+
1 row in set (0.02 sec)

mysql> LOAD DATA LOCAL INFILE 'C:/Users/Asus/Downloads/world_population.csv'
  -> INTO TABLE world_population
  -> FIELDS TERMINATED BY ','
  -> ENCLOSED BY '"'
  -> LINES TERMINATED BY '\n'
  -> IGNORE 1 ROWS;
Query OK, 234 rows affected (0.03 sec)
Records: 234 Deleted: 0 Skipped: 0 Warnings: 0
```

QUERIES EXECUTION:-

Query 1: Population Density

```
mysql> SELECT
  ->   `Country/Territory`,
  ->   `2022 Population` / `Area (km²)` AS population_density
  -> FROM world_population;
```

Country/Territory	population_density
Afghanistan	63.0587
Albania	98.8702
Algeria	18.8531
American Samoa	222.4774
Andorra	170.5641
Angola	28.5466
Anguilla	174.2527
Antigua and Barbuda	212.1335
Argentina	16.3683
Armenia	93.4831
Aruba	591.3611
Australia	3.4032
Austria	106.5877
Azerbaijan	119.6082
Bahamas	29.4043
Bahrain	1924.4876
Bangladesh	1160.0350
Barbados	654.9651
Belarus	45.9295
Belgium	381.8111
Belize	17.6466
Benin	118.5635
Bermuda	1188.5926
Bhutan	20.3796
Bolivia	11.1272
Bosnia and Herzegovina	63.1437
Botswana	4.5194
Brazil	25.2841
British Virgin Islands	207.3179
Brunei	77.8841
Bulgaria	61.1654
Burkina Faso	83.0641
Burundi	463.0874
Cambodia	92.6221
Cameroon	58.7128
Canada	3.8513
Cape Verde	147.0739
Cayman Islands	260.2500
Central African Republic	8.9555

Slovakia	115.0856
Slovenia	104.5649
Solomon Islands	25.0648
Somalia	27.5971
South Africa	49.0517
South Korea	517.0722
South Sudan	17.6091
Spain	93.9969
Sri Lanka	332.7563
Sudan	24.8529
Suriname	3.7727
Sweden	23.4276
Switzerland	211.7157
Syria	119.4797
Taiwan	660.1662
Tajikistan	69.5513
Tanzania	69.3034
Thailand	139.7276
Timor-Leste	90.1772
Togo	155.8281
Tokelau	155.9167
Tonga	143.0495
Trinidad and Tobago	298.4491
Tunisia	75.5218
Turkey	108.9145
Turkmenistan	13.1751
Turks and Caicos Islands	48.2099
Tuvalu	435.0769
Uganda	195.6100
Ukraine	65.7858
United Arab Emirates	112.9322
United Kingdom	277.9289
United States	36.0935
United States Virgin Islands	286.6427
Uruguay	18.9069
Uzbekistan	77.3975
Vanuatu	26.8061
Vatican City	510.0000
Venezuela	30.8820
Vietnam	296.4472
Wallis and Futuna	81.4930
Western Sahara	2.1654
Yemen	63.8232
Zambia	26.5976
Zimbabwe	41.7665

234 rows in set (0.00 sec)

Purpose

Calculates how many people live in one square kilometer.

Formula

Population Density = Population ÷ Area

Functions Used

- / Division Operator
- ROUND() rounds result to 2 decimal places.

Query 2: Total Population of All Countries

```
mysql> SELECT
  -> SUM(`2022 Population`) AS total_population
  -> FROM world_population;

+-----+
| total_population |
+-----+
|      7973413042 |
+-----+
1 row in set (0.00 sec)
```

Purpose

Finds the combined population of all countries.

Formula

Total Population =

Sum of all country populations

Function Used

SUM()

Aggregate function that adds all values.

Query 3: Difference Between Largest and Second Largest Population

```
mysql> SELECT
-> (
->     SELECT `2022 Population`
->     FROM world_population
->     ORDER BY `2022 Population` DESC
->     LIMIT 1
-> )
-> -
-> (
->     SELECT `2022 Population`
->     FROM world_population
->     ORDER BY `2022 Population` DESC
->     LIMIT 1 OFFSET 1
-> )
-> AS population_difference;
+-----+
| population_difference |
+-----+
|          8714164     |
+-----+
1 row in set (0.00 sec)
```

Purpose

Measures the gap between the most populated and second most populated country.

Steps

- 1. Find largest population.***
- 2. Find second largest population.***
- 3. Subtract them.***

Concepts:-

- Subquery ,ORDER BY LIMIT OFFSET***

Q4 Population Increased By 10%

Purpose

Predicts population after a 10% increase.

Formula

Future Population =

Current Population × 1.10

```

ysql> SELECT
-> `Country/Territory`,
-> `2022 Population`,
-> ROUND(C 2022 Population*1.10) AS increased_population
-> FROM world_population;

```

Country/Territory	2022 Population	increased_population
Afghanistan	41128771	45241648
Albania	2842321	3126553
Algeria	4498325	49393548
American Samoa	44273	48700
Andorra	79824	87806
Angola	35588987	39147886
Anguilla	15887	17443
Antigua and Barbuda	93763	103139
Argentina	45510318	50061350
Armenia	2780469	3058516
Aruba	106445	117090
Australia	261777413	28795154
Austria	8939617	9833579
Azerbaijan	10358074	11393881
Bahamas	409984	450982
Bahrain	1472233	1619456
Bangladesh	171186372	188305009
Barbados	281635	309799
Belarus	9534954	10488449
Belgium	11655930	12821523
Belize	405272	445799
Benin	13352864	14688150
Bermuda	64184	70602
Bhutan	782455	860701
Bolivia	12224110	13446521
Bosnia and Herzegovina	3233526	3556879
Botswana	2630296	2893326
Brazil	215313498	236844848
British Virgin Islands	31305	34436
Brunei	449002	493902
Bulgaria	6781953	7460148
Burkina Faso	22673762	24941138
Burundi	12809576	14178534
Cambodia	16767842	18444624
Cameroon	27914536	30705990
Canada	38454327	42299760
Cape Verde	593449	652464
Cayman Islands	68706	75577
Central African Republic	5579144	6137058
Chad	17723315	19495647
Chile	19603723	21564106
China	1425887337	1568476071
Colombia	51874024	57061426
Comoros	836774	920451
Cook Islands	17011	18712
Costa Rica	5180829	5698912
Croatia	4030358	4433394
Cuba	11212191	12333410
Curaçao	191163	210279
Cyprus	1251488	1376637
Czech Republic	10493986	11543385
Denmark	588261	6470487
Djibouti	1120849	1232934
Dominica	72737	80011
Dominican Republic	11228821	12351703
DR Congo	90810212	100911233
Ecuador	18001000	19801100
Egypt	110990103	122089113
El Salvador	6336392	6970031
Equatorial Guinea	1674908	1842399
Eritrea	3684032	4052435
Estonia	1326062	1458668

Poland	39857145	43842860
Portugal	10270865	11297952
Puerto Rico	3252407	3577648
Qatar	2685122	2964634
Republic of the Congo	5970424	6567466
Reunion	974052	1071457
Romania	19659267	21625194
Russia	144713314	159184645
Rwanda	13776698	15154368
Saint Barthelemy	10967	12064
Saint Kitts and Nevis	47657	52423
Saint Lucia	179857	197843
Saint Martin	31791	34970
Saint Pierre and Miquelon	5862	6448
Saint Vincent and the Grenadines	103948	114343
Samoa	222382	244620
San Marino	33660	37026
Sao Tome and Principe	227380	250118
Saudi Arabia	36408820	40049702
Senegal	17316449	19048094
Serbia	7221365	7943502
Seychelles	107118	117830
Sierra Leone	8605718	9466290
Singapore	5975689	6573258
Sint Maarten	44175	48593
Slovakia	5643453	6207798
Slovenia	2119844	2331828
Solomon Islands	724273	796700
Somalia	17597511	19357262
South Africa	59893885	65883274
South Korea	51815810	56997391
South Sudan	10913164	12004480
Spain	47558630	52314493
Sri Lanka	21832143	24015357
Sudan	46874204	51561624
Suriname	618040	679844
Sweden	10549347	11604282
Switzerland	8740472	9614519
Syria	22125249	24337774
Taiwan	23893394	26282723
Tajikistan	9952787	10948066
Tanzania	65497748	72047523
Thailand	71697030	78866733
Timor-Leste	1341296	1475426
Togo	8848699	9733569
Tokelau	1871	2058
Tonga	106858	117544
Trinidad and Tobago	1531044	1684148
Tunisia	12356117	13591729
Turkey	85341241	93875365
Turkmenistan	6430770	7073847
Turks and Caicos Islands	45703	50273
Tuvalu	11312	12443
Uganda	47249585	51974544
Ukraine	39701739	43671913
United Arab Emirates	9441129	10385242
United Kingdom	67508936	74259830
United States	338289857	372118843
United States Virgin Islands	99465	109412
Uruguay	3422793	3765073
Uzbekistan	34627652	38090417
Vanuatu	326740	359414
Vatican City	510	561
Venezuela	28301606	31131806
Vietnam	98186856	108005542
Wallis and Futuna	11572	12729
Western Sahara	575986	633585
Yemen	33996614	37066275
Zambia	20017675	22019443
Zimbabwe	16320537	17952591

34 rows in set (0.01 sec)

Q5 Area Conversion To Square Miles

```
mysql> SELECT
-> `Country/Territory`,
-> ROUND(`Area (km²)`*0.386102,2) AS area_sq_miles
-> FROM world_population;
```

Country/Territory	area_sq_miles
Afghanistan	251827.31
Albania	11099.66
Algeria	919594.06
American Samoa	76.83
Andorra	180.70
Angola	481353.36
Anguilla	35.14
Antigua and Barbuda	170.66
Argentina	1073518.00
Armenia	11483.83
Aruba	69.50
Australia	2969905.85
Austria	32382.76
Azerbaijan	33436.43
Bahamas	5383.42
Bahrain	295.37
Bangladesh	56977.07
Barbados	166.02
Belarus	80154.78
Belgium	11786.92
Belize	8867.22
Benin	43483.58
Bermuda	20.85
Bhutan	14824.00
Bolivia	424164.32
Bosnia and Herzegovina	19771.90
Botswana	224711.36
Brazil	3287954.67
British Virgin Islands	58.30
Brunei	2225.88
Bulgaria	42810.60
Burkina Faso	105393.10
Burundi	10746.76
Cambodia	69897.98
Cameroon	183569.11
Canada	3855101.06
Cape Verde	1557.15
Cayman Islands	101.93
Central African Republic	240535.37
Chad	495754.97
Chile	291932.49
China	3747877.06
Colombia	440831.19
Comoros	718.92
Cook Islands	91.12
Costa Rica	19729.81
Croatia	21851.06
Cuba	42426.43
Curacao	171.43
Cyprus	3571.83
Czech Republic	30449.93
Denmark	16638.68
Djibouti	8957.57
Dominica	289.96
Dominican Republic	18791.97
DR Congo	905354.36
Ecuador	106888.86
Egypt	387047.95
El Salvador	8123.97
Equatorial Guinea	10830.55

Purpose

Converts area from square kilometers to square miles.

Formula

$1 \text{ km}^2 = 0.386102 \text{ square miles}$

Puerto Rico	3424.72
Qatar	4473.38
Republic of the Congo	132046.88
Reunion	969.50
Romania	92043.24
Russia	6601665.43
Rwanda	10169.15
Saint Barthelemy	8.11
Saint Kitts and Nevis	100.77
Saint Lucia	237.84
Saint Martin	20.46
Saint Pierre and Miquelon	93.44
Saint Vincent and the Grenadines	150.19
Samoa	1097.30
San Marino	23.55
Sao Tome and Principe	372.20
Saudi Arabia	829999.61
Senegal	75954.76
Serbia	34116.36
Seychelles	174.52
Sierra Leone	27698.96
Singapore	274.13
Sint Maarten	13.13
Slovakia	18933.28
Slovenia	7827.45
Solomon Islands	11156.80
Somalia	246200.64
South Africa	471444.83
South Korea	38691.28
South Sudan	239284.78
Spain	195364.52
Sri Lanka	25332.15
Sudan	728214.63
Suriname	63251.23
Sweden	173859.80
Switzerland	15939.83
Syria	71498.37
Taiwan	13974.19
Tajikistan	55251.20
Tanzania	364899.98
Thailand	198116.66
Timor-Leste	5742.88
Togo	21924.80
Tokelau	4.63
Tonga	288.42
Trinidad and Tobago	1980.70
Tunisia	63170.15
Turkey	302534.86
Turkmenistan	188456.39
Turks and Caicos Islands	366.02
Tuvalu	10.04
Uganda	93262.94
Ukraine	233012.56
United Arab Emirates	32278.13
United Kingdom	93784.18
United States	3618783.47
United States Virgin Islands	1133.98
Uruguay	69897.59
Uzbekistan	172742.03
Vanuatu	4706.20
Vatican City	0.39
Venezuela	353841.25
Vietnam	127881.62
Wallis and Futuna	54.83
Western Sahara	102703.13
Yemen	203849.50
Zambia	290585.00
Zimbabwe	150872.06

4 rows in set (0.00 sec)

Q6 Population In Millions

```
mysql> SELECT
-> 'Country/Territory',
-> '2022 Population' DIV 1000000 AS population_millions
-> FROM world_population;
```

Country/Territory	population_millions
Afghanistan	41
Albania	2
Algeria	44
American Samoa	0
Andorra	0
Angola	35
Anguilla	0
Antigua and Barbuda	0
Argentina	45
Armenia	2
Aruba	0
Australia	26
Austria	8
Azerbaijan	10
Bahamas	0
Bahrain	1
Bangladesh	171
Barbados	0
Belarus	9
Belgium	11
Belize	0
Benin	13
Bermuda	0
Bhutan	0
Bolivia	12
Bosnia and Herzegovina	3
Botswana	2
Brazil	215
British Virgin Islands	0
Brunei	0
Bulgaria	6
Burkina Faso	22
Burundi	12
Cambodia	16
Cameroon	27
Canada	38
Cape Verde	0
Cayman Islands	0
Central African Republic	5
Chad	17
Chile	19
China	1425
Colombia	51
Comoros	0
Cook Islands	0
Costa Rica	5
Croatia	4
Cuba	11
Curacao	0
Cyprus	1
Czech Republic	10
Denmark	5
Djibouti	1
Dominica	0
Dominican Republic	11
DR Congo	99
Ecuador	18
Egypt	110

Purpose

Shows population in million units.

Operator Used

DIV

Integer division

Puerto Rico	3
Qatar	2
Republic of the Congo	5
Reunion	0
Romania	19
Russia	144
Rwanda	13
Saint Barthelemy	0
Saint Kitts and Nevis	0
Saint Lucia	0
Saint Martin	0
Saint Pierre and Miquelon	0
Saint Vincent and the Grenadines	0
Samoa	0
San Marino	0
Sao Tome and Principe	0
Saudi Arabia	36
Senegal	17
Serbia	7
Seychelles	0
Sierra Leone	8
Singapore	5
Sint Maarten	0
Slovakia	5
Slovenia	2
Solomon Islands	0
Somalia	17
South Africa	59
South Korea	51
South Sudan	10
Spain	47
Sri Lanka	21
Sudan	46
Suriname	0
Sweden	10
Switzerland	8
Syria	22
Taiwan	23
Tajikistan	9
Tanzania	65
Thailand	71
Timor-Leste	1
Togo	8
Tokelau	0
Tonga	0
Trinidad and Tobago	1
Tunisia	12
Turkey	85
Turkmenistan	6
Turks and Caicos Islands	0
Tuvalu	0
Uganda	47
Ukraine	39
United Arab Emirates	9
United Kingdom	67
United States	338
United States Virgin Islands	0
Uruguay	3
Uzbekistan	34
Vanuatu	0
Vatican City	0
Venezuela	28
Vietnam	98
Wallis and Futuna	0
Western Sahara	0
Yemen	33
Zambia	20
Zimbabwe	16

234 rows in set (0.00 sec)

Q7 Population Multiple Of One Million(MOD() operator)

```
mysql> SELECT
-> `Country/Territory`,
-> `2022 Population`,
-> MOD(`2022 Population`,1000000) AS remainder
-> FROM world_population
-> LIMIT 20;
```

Country/Territory	2022 Population	remainder
Afghanistan	41128771	128771
Albania	2842321	842321
Algeria	44903225	903225
American Samoa	44273	44273
Andorra	79824	79824
Angola	35588987	588987
Anguilla	15857	15857
Antigua and Barbuda	93763	93763
Argentina	45510318	510318
Armenia	2780469	780469
Aruba	106445	106445
Australia	26177413	177413
Austria	8939617	939617
Azerbaijan	10358074	358074
Bahamas	409984	409984
Bahrain	1472233	472233
Bangladesh	171186372	186372
Barbados	281635	281635
Belarus	9534954	534954
Belgium	11655930	655930

20 rows in set (0.00 sec)

Purpose

**Finds populations exactly
divisible by 1 million.**

Operator Used

MOD()

Returns remainder.

Q8 Share Of World Population

```
mysql> SELECT
-> `Country/Territory`,
-> ROUND(
-> (`2022 Population`/
-> (SELECT SUM(`2022 Population`) FROM world_population))*100
-> ,2) AS population_share_percent
-> FROM world_population;
```

Country/Territory	population_share_percent
Afghanistan	0.52
Albania	0.04
Algeria	0.56
American Samoa	0.00
Andorra	0.00
Angola	0.45
Anguilla	0.00
Antigua and Barbuda	0.00
Argentina	0.57
Armenia	0.03
Aruba	0.00
Australia	0.33
Austria	0.11
Azerbaijan	0.13
Bahamas	0.01
Bahrain	0.02
Bangladesh	2.15
Barbados	0.00
Belarus	0.12
Belgium	0.15
Belize	0.01
Benin	0.17
Bermuda	0.00
Bhutan	0.01
Bolivia	0.15
Bosnia and Herzegovina	0.04
Botswana	0.03
Brazil	2.70
British Virgin Islands	0.00
Brunei	0.01
Bulgaria	0.09
Burkina Faso	0.28
Burundi	0.16
Cambodia	0.21
Cameroon	0.35
Canada	0.48
Cape Verde	0.01
Cayman Islands	0.00
Central African Republic	0.07
Chad	0.22
Chile	0.25
China	17.88
Colombia	0.65
Comoros	0.01
Cook Islands	0.00
Costa Rica	0.06

Puerto Rico	0.04
Qatar	0.03
Republic of the Congo	0.07
Reunion	0.01
Romania	0.25
Russia	1.81
Rwanda	0.17
Saint Barthelemy	0.08
Saint Kitts and Nevis	0.00
Saint Lucia	0.00
Saint Martin	0.00
Saint Pierre and Miquelon	0.00
Saint Vincent and the Grenadines	0.00
Samoa	0.00
San Marino	0.00
Sao Tome and Principe	0.00
Saudi Arabia	0.46
Senegal	0.22
Serbia	0.09
Seychelles	0.00
Sierra Leone	0.11
Singapore	0.07
Sint Maarten	0.00
Slovakia	0.07
Slovenia	0.03
Solomon Islands	0.01
Somalia	0.22
South Africa	0.75
South Korea	0.65
South Sudan	0.14
Spain	0.60
Sri Lanka	0.27
Sudan	0.59
Suriname	0.01
Sweden	0.13
Switzerland	0.11
Syria	0.28
Taiwan	0.30
Tajikistan	0.12
Tanzania	0.82
Thailand	0.90
Timor-Leste	0.02
Togo	0.11
Tokelau	0.00
Tonga	0.00
Trinidad and Tobago	0.02
Tunisia	0.15
Turkey	1.07
Turkmenistan	0.08
Turks and Caicos Islands	0.00
Tuvalu	0.00
Uganda	0.59
Ukraine	0.50
United Arab Emirates	0.12
United Kingdom	0.85
United States	4.24
United States Virgin Islands	0.00
Uruguay	0.04
Uzbekistan	0.43
Vanuatu	0.00
Vatican City	0.00
Venezuela	0.35
Vietnam	1.23
Wallis and Futuna	0.00
Western Sahara	0.01
Yemen	0.42
Zambia	0.25
Zimbabwe	0.20

234 rows in set (0.00 sec)

Purpose

Calculates each country's contribution to total world population.

Formula

Country Population

×100

World Population

Concepts

• **Subquery**

- **SUM()**
- **Percentage calculation**

Q9 Average Population Per Continent

```
mysql> SELECT
-> Continent,
-> ROUND(AVG(`2022 Population`),0) AS average_population
-> FROM world_population
-> GROUP BY Continent;
```

Continent	average_population
Asia	94427665
Europe	14862951
Africa	25030367
Oceania	1958198
North America	15007403
South America	31201186

6 rows in set (0.01 sec)

Purpose

Finds average country population within each continent.

Function

AVG() Concept; GROUP BY:- Groups countries by continent

Q10 Population Projection After 10 Years

```
mysql> SELECT
-> `Country/Territory`,
-> ROUND(`2022 Population`*POWER(1.02,10))
-> AS projected_population
-> FROM world_population;
```

Country/Territory	projected_population
Afghanistan	50135742
Albania	3464773
Algeria	54723781
American Samoa	53969
Andorra	97305
Angola	43382777
Anguilla	19330
Antigua and Barbuda	114297
Argentina	55476824
Armenia	3389376
Aruba	129756
Australia	31910120
Austria	1794649
Azerbaijan	12626434
Bahamas	499768
Bahrain	1794649
Bangladesh	208675232
Barbados	343311
Belarus	11623056
Belgium	14208514
Belize	494024
Benin	16277067
Bermuda	78240
Bhutan	953808
Bolivia	14901122
Bosnia and Herzegovina	3941630
Botswana	3206316
Brazil	262465953
British Virgin Islands	38161
Brunei	547331
Bulgaria	8267163
Burkina Faso	27639189
Burundi	15712321
Cambodia	20439906
Cameroon	34027664
Canada	46875619
Cape Verde	723045
Cayman Islands	83752
Central African Republic	6880945
Chad	21604622
Chile	23896841
China	1738148707
Colombia	63234146
Comoros	1020023
Cook Islands	20736
Costa Rica	6315402
Croatia	4912984
Cuba	13667598
Curacao	233027
Cyprus	1525557
Czech Republic	12792110
Denmark	7170443
Djibouti	1366399
Dominica	88666
Dominican Republic	13687870
DR Congo	120692896
Ecuador	21943119
Egypt	135296316
El Salvador	7724026
Equatorial Guinea	2041704
Eritrea	4490814
Estonia	161862
Eswatini	1464829
Ethiopia	150399439

*Purpose Projects
population after
10 years
assuming 2%
annual growth.*

*Formula :-Future
Population =*

*Present × (1+r)ⁿ
Where:*

r = growth rate

n = years

Function

UsedPOWER()

Puerto Rico	3964666
Qatar	3295339
Republic of the Congo	7277314
Reunion	1187364
Romania	23964537
Russia	176404722
Rwanda	16793718
Saint Barthelemy	13369
Saint Kitts and Nevis	58094
Saint Lucia	219245
Saint Martin	38753
Saint Pierre and Miquelon	7146
Saint Vincent and the Grenadines	126712
Samoa	271082
San Marino	41031
Sao Tome and Principe	277175
Saudi Arabia	44382148
Senegal	21108655
Serbia	8802804
Seychelles	130576
Sierra Leone	10490322
Singapore	7284332
Sint Maarten	53849
Slovakia	6879238
Slovenia	2584078
Solomon Islands	882885
Somalia	21451268
South Africa	73010312
South Korea	63163183
South Sudan	13303086
Spain	57973705
Sri Lanka	26613260
Sudan	57139393
Suriname	753387
Sweden	12859595
Switzerland	10654587
Syria	26970555
Taiwan	29125914
Tajikistan	12132392
Tanzania	79841389
Thailand	87398280
Timor-Leste	1635032
Togo	10786515
Tokelau	2281
Tonga	130259
Trinidad and Tobago	1866334
Tunisia	15062038
Turkey	104030497
Turkmenistan	7839073
Turks and Caicos Islands	55712
Tuvalu	13789
Uganda	57596980
Ukraine	48396198
United Arab Emirates	11508684
United Kingdom	82293016
United States	412373448
United States Virgin Islands	121247
Uruguay	4172367
Uzbekistan	42210915
Vanuatu	398294
Vatican City	622
Venezuela	34499610
Vietnam	119680230
Wallis and Futuna	14106
Western Sahara	702124
Yemen	41075984
Zambia	24401434
Zimbabwe	19894644

234 rows in set (0.01 sec)

Q11 Population Modulus Area

```
mysql> SELECT
-> `Country/Territory`,
-> MOD(`2022 Population`,`Area (km²)`) AS remainder_value
-> FROM world_population;
```

Country/Territory	remainder_value
Afghanistan	38281
Albania	25017
Algeria	2031887
American Samoa	95
Andorra	264
Angola	681387
Anguilla	23
Antigua and Barbuda	59
Argentina	1023918
Armenia	14370
Aruba	65
Australia	3101341
Austria	40291
Azerbaijan	52674
Bahamas	5637
Bahrain	273
Bangladesh	5172
Barbados	415
Belarus	192954
Belgium	24762
Belize	14850
Benin	63468
Bermuda	32
Bhutan	14575
Bolivia	139719
Bosnia and Herzegovina	7359
Botswana	302296
Brazil	2419323
British Virgin Islands	86
Brunei	5097
Bulgaria	18334
Burkina Faso	17501
Burundi	2434
Cambodia	112622
Cameroon	338900
Canada	8500317
Cape Verde	298
Cayman Islands	86
Central African Republic	595272
Chad	1031315
Chile	701183
China	8671031
Colombia	495364
Comoros	736
Cook Islands	19
Costa Rica	19729
Croatia	12184
Cuba	4023
Curacao	243
Cyprus	2603
Czech Republic	4941
Denmark	21477
Djibouti	7249
Dominica	641
Dominican Republic	34491
DR Congo	526176
Ecuador	6335
Egypt	720603
El Salvador	3051
Equatorial Guinea	19399
Eritrea	38432
Estonia	14479
Eswatini	3554

Purpose

*Demonstrates
modulus
operation
between two
columns.*

Formula

*Population MOD
Area*

Puerto Rico	5987
Qatar	7170
Republic of the Congo	156424
Reunion	2295
Romania	111205
Russia	7927378
Rwanda	1924
Saint Barthelemy	5
Saint Kitts and Nevis	155
Saint Lucia	601
Saint Martin	44
Saint Pierre and Miquelon	54
Saint Vincent and the Grenadines	85
Samoa	706
San Marino	49
Sao Tome and Principe	840
Saudi Arabia	2013780
Senegal	4913
Serbia	64124
Seychelles	446
Sierra Leone	68658
Singapore	329
Sint Maarten	9
Slovakia	4198
Slovenia	11452
Solomon Islands	1873
Somalia	380772
South Africa	63072
South Korea	7240
South Sudan	377499
Spain	501374
Sri Lanka	49623
Sudan	1608572
Suriname	126580
Sweden	192562
Switzerland	29548
Syria	88829
Taiwan	6014
Tajikistan	78887
Tanzania	286745
Thailand	373350
Timor-Leste	2636
Togo	47024
Tokelau	11
Tonga	37
Trinidad and Tobago	2304
Tunisia	85367
Turkey	716545
Turkmenistan	85470
Turks and Caicos Islands	199
Tuvalu	2
Uganda	147335
Ukraine	474239
United Arab Emirates	77929
United Kingdom	225636
United States	875897
United States Virgin Islands	223
Uruguay	164182
Uzbekistan	177852
Vanuatu	9826
Vatican City	0
Venezuela	808346
Vietnam	148104
Wallis and Futuna	70
Western Sahara	43308
Yemen	434630
Zambia	449763
Zimbabwe	299500

234 rows in set (0.00 sec)

Q12 Population Bucket

```
234 rows in set (0.00 sec)

mysql> SELECT
  -> 'Country/Territory',
  -> FLOOR('2022 Population'/10000000) AS population_bucket
  -> FROM world_population;
```

Country/Territory	population_bucket
Afghanistan	4
Albania	0
Algeria	4
American Samoa	0
Andorra	0
Angola	3
Anguilla	0
Antigua and Barbuda	0
Argentina	4
Armenia	0
Aruba	0
Australia	2
Austria	0
Azerbaijan	1
Bahamas	0
Bahrain	0
Bangladesh	17
Barbados	0
Belarus	0
Belgium	1
Belize	0
Benin	1
Bermuda	0
Bhutan	0
Bolivia	1
Bosnia and Herzegovina	0
Botswana	0
Brazil	21
British Virgin Islands	0
Brunei	0
Bulgaria	0
Burkina Faso	2
Burundi	1
Cambodia	1
Cameroon	2
Canada	3
Cape Verde	0
Cayman Islands	0
Central African Republic	0
Chad	1
Chile	1
China	142
Colombia	5
Comoros	0
Cook Islands	0
Costa Rica	0
Croatia	0
Cuba	1
Curacao	0
Cyprus	0
Czech Republic	1
Denmark	0
Djibouti	0
Dominica	0
Dominican Republic	1
DR Congo	9
Ecuador	1
Egypt	11
El Salvador	0
Equatorial Guinea	0
Eritrea	0
Puerto Rico	0
Qatar	0
Republic of the Congo	0
Reunion	1
Romania	14
Russia	1
Rwanda	0
Saint Barthelemy	0
Saint Kitts and Nevis	0
Saint Lucia	0
Saint Martin	0
Saint Pierre and Miquelon	0
Saint Vincent and the Grenadines	0
Samoa	0
San Marino	0
Sao Tome and Principe	0
Saudi Arabia	3
Senegal	1
Serbia	0
Seychelles	0
Sierra Leone	0
Singapore	0
Sint Maarten	0
Slovakia	0
Slovenia	0
Solomon Islands	0
Somalia	1
South Africa	5
South Korea	5
South Sudan	1
Spain	14
Sri Lanka	4
Sudan	4
Suriname	0
Sweden	1
Switzerland	1
Syria	2
Taiwan	2
Tajikistan	0
Tanzania	4
Thailand	7
Timor-Leste	0
Togo	0
Tokelau	0
Tonga	0
Trinidad and Tobago	0
Tunisia	1
Turkey	28
Turkmenistan	0
Turks and Caicos Islands	0
Tuvalu	0
Uganda	4
Ukraine	0
United Arab Emirates	0
United Kingdom	6
United States	33
United States Virgin Islands	0
Uruguay	0
Uzbekistan	3
Vanuatu	0
Vatican City	0
Venezuela	0
Vietnam	9
Wallis and Futuna	0
Western Sahara	0
Yemen	0
Zambia	1
Zimbabwe	1

```
34 rows in set (0.00 sec)
```

Purpose

Categorizes
countries into population
groups.

Function

FLOOR()

Removes decimal
part.

Q13 Population In Million Rounded

```
mysql> SELECT
-> `Country/Territory`,
-> ROUND(`2022 Population`/1000000,2)
-> AS population_in_million
-> FROM world_population;
```

Country/Territory	population_in_million
Afghanistan	41.13
Albania	2.84
Algeria	44.90
American Samoa	0.04
Andorra	0.08
Angola	35.59
Anguilla	0.02
Antigua and Barbuda	0.09
Argentina	45.51
Armenia	2.78
Aruba	0.11
Australia	26.18
Austria	8.94
Azerbaijan	10.36
Bahamas	0.41
Bahrain	1.47
Bangladesh	171.19
Barbados	0.28
Belarus	9.53
Belgium	11.66
Belize	0.41
Benin	13.35
Bermuda	0.06
Bhutan	0.78
Bolivia	12.22
Bosnia and Herzegovina	3.23
Botswana	2.63
Brazil	215.31
British Virgin Islands	0.03
Brunei	0.45
Bulgaria	6.78
Burkina Faso	22.67
Burundi	12.89
Cambodia	16.77
Cameroon	27.91
Canada	38.45
Cape Verde	0.59
Cayman Islands	0.07
Central African Republic	5.58
Chad	17.72
Chile	19.60
China	1425.89
Colombia	51.87
Comoros	0.84
Cook Islands	0.02
Costa Rica	5.18
Croatia	4.03
Cuba	11.21
Curacao	0.10
Cyprus	1.25
Czech Republic	10.49
Denmark	5.88
Djibouti	1.12
Dominica	0.07
Dominican Republic	11.23
DR Congo	99.01
Ecuador	18.00
Egypt	110.99
El Salvador	6.34
Equatorial Guinea	1.67
Eritrea	3.68

Qatar	2.70
Republic of the Congo	5.97
Reunion	0.97
Romania	19.66
Russia	144.71
Rwanda	13.78
Saint Barthelemy	0.01
Saint Kitts and Nevis	0.05
Saint Lucia	0.18
Saint Martin	0.03
Saint Pierre and Miquelon	0.01
Saint Vincent and the Grenadines	0.10
Samoa	0.22
San Marino	0.03
Sao Tome and Principe	0.23
Saudi Arabia	36.41
Senegal	17.32
Serbia	7.22
Seychelles	0.11
Sierra Leone	8.61
Singapore	5.95
Sint Maarten	0.04
Slovakia	5.64
Slovenia	2.12
Solomon Islands	0.72
Somalia	17.60
South Africa	59.89
South Korea	51.82
South Sudan	10.91
Spain	47.56
Sri Lanka	21.83
Sudan	46.87
Suriname	0.62
Sweden	10.55
Switzerland	8.74
Syria	22.13
Taiwan	23.89
Tajikistan	9.95
Tanzania	65.50
Thailand	71.70
Timor-Leste	1.34
Togo	8.85
Tokelau	0.00
Tonga	0.11
Trinidad and Tobago	1.53
Tunisia	12.36
Turkey	85.34
Turkmenistan	6.43
Turks and Caicos Islands	0.05
Tuvalu	0.01
Uganda	47.25
Ukraine	39.70
United Arab Emirates	9.44
United Kingdom	67.51
United States	338.29
United States Virgin Islands	0.10
Uruguay	3.42
Uzbekistan	34.63
Vanuatu	0.33
Vatican City	0.00
Venezuela	28.30
Vietnam	98.19
Wallis and Futuna	0.01
Western Sahara	0.58
Yemen	33.70
Zambia	20.00
Zimbabwe	16.22

234 rows in set (0.00 sec)

Purpose

**Displays
population in a
more readable
form.**

Example

1,428,627,663

=1428.63 million

Q14 Area Ratio Compared To Largest Country

```
mysql> SELECT
-> `Country/Territory`,
-> ROUND(`2022 Population`/1000000,2)
-> AS population_in_million
-> FROM world_population;
```

Country/Territory	population_in_million
Afghanistan	41.13
Albania	2.84
Algeria	44.90
American Samoa	0.04
Andorra	0.08
Angola	35.89
Anguilla	0.02
Antigua and Barbuda	0.09
Argentina	45.51
Armenia	2.78
Aruba	0.11
Australia	26.18
Austria	8.94
Azerbaijan	10.36
Bahamas	0.41
Bahrain	1.47
Bangladesh	171.19
Barbados	0.28
Belarus	9.53
Belgium	11.66
Belize	0.41
Benin	13.35
Bermuda	0.06
Bhutan	0.78
Bolivia	12.22
Bosnia and Herzegovina	3.23
Botswana	2.63
Brazil	215.31
British Virgin Islands	0.03
Brunei	0.45
Bulgaria	6.78
Burkina Faso	22.67
Burundi	11.89
Cambodia	16.77
Cameroon	27.91
Canada	38.45
Cape Verde	0.59
Cayman Islands	0.07

Puerto Rico	1927.65
Qatar	1475.77
Republic of the Congo	49.99
Reunion	6809.34
Romania	71.72
Russia	1.00
Rwanda	649.19
Saint Barthelemy	814202.00
Saint Kitts and Nevis	65510.51
Saint Lucia	27756.89
Saint Martin	322608.34
Saint Pierre and Miquelon	70653.89
Saint Vincent and the Grenadines	43954.35
Samoa	6016.27
San Marino	280299.05
Sao Tome and Principe	17736.77
Saudi Arabia	7.95
Senegal	86.92
Serbia	193.50
Seychelles	37827.97
Sierra Leone	238.34
Singapore	24082.03
Sint Maarten	502889.47
Slovakia	348.68
Slovenia	843.40
Solomon Islands	521.72
Somalia	26.81
South Africa	14.00
South Korea	170.62
South Sudan	27.59
Spain	33.79
Sri Lanka	260.60
Sudan	9.07
Suriname	104.37
Sweden	37.97
Switzerland	414.16
Syria	92.33
Taiwan	472.42
Tajikistan	119.48
Tanzania	18.09
Thailand	33.32
Timor-Leste	1149.54
Togo	301.10
Tokelau	1424853.50
Tonga	22889.21
Trinidad and Tobago	3332.90
Tunisia	104.51
Turkey	21.82
Turkmenistan	35.03
Turks and Caicos Islands	18036.12
Tuvalu	657624.69
Uganda	70.79
Ukraine	28.33
United Arab Emirates	204.52
United Kingdom	70.39
United States	1.82
United States Virgin Islands	49274.47
Uruguay	94.45
Uzbekistan	38.22
Vanuatu	1402.76
Vatican City	17098242.00
Venezuela	18.66
Vietnam	51.62
Wallis and Futuna	120410.15
Western Sahara	64.28
Yemen	32.38
Zambia	22.72
Zimbabwe	43.76

234 rows in set (0.01 sec)

Purpose

Compares each country's area with the largest country.

Formula

Largest Area/

Country Area

Concepts

- MAX()
- Subquery

Q15 Five Year Population Projection Using Growth Rate

```
mysql> SELECT
-> 'Country/Territory',
-> ROUND(
-> 2022 Population*
-> POWER('Growth Rate',5)
-> )
-> AS projected_5_year_population
-> FROM world_population;
```

Country/Territory	projected_5_year_population
Afghanistan	46692541
Albania	2781734
Algeria	48708058
American Samoa	40656
Andorra	83896
Angola	41858884
Anguilla	16387
Antigua and Barbuda	96514
Argentina	46705956
Armenia	2728626
Aruba	105967
Australia	27499107
Austria	9029371
Azerbaijan	10587536
Bahamas	420546
Bahrain	1517687
Bangladesh	180632276
Barbados	283754
Belarus	922340
Belgium	11879082
Belize	432522
Benin	15285239
Bermuda	64184
Bhutan	807816
Bolivia	12975372
Bosnia and Herzegovina	3053370
Botswana	2850266
Brazil	220311479
British Virgin Islands	32239
Brunei	467484
Bulgaria	6295147
Burkina Faso	25766103
Burundi	14726205
Cambodia	17693076
Cameroon	31783525
Canada	39977625
Cape Verde	620018
Cayman Islands	71641
Central African Republic	6232636
Chad	20706258
Chile	20168845
China	1425887337
Colombia	53688546
Comoros	916643
Cook Islands	17054
Costa Rica	5316939
Croatia	3885382
Cuba	10995252
Curacao	195309
Cyprus	1288845
Czech Republic	10410302
Denmark	6024797
Djibouti	1200352
Dominica	74388
Dominican Republic	11801604
DR Congo	116179711
Ecuador	19050719
Egypt	120039809
El Salvador	6448858
Equatorial Guinea	1892233
Eritrea	4019841

Purpose

**Projects population
after 5 years using actual
growth rate.**

Formula

**Population ×
GrowthRate^5**

Puerto Rico	3234558
Qatar	2730341
Republic of the Congo	6692611
Reunion	1014648
Romania	21398611
Russia	142770205
Rwanda	15465793
Saint Barthelemy	11515
Saint Kitts and Nevis	47920
Saint Lucia	180848
Saint Martin	31020
Saint Pierre and Miquelon	5787
Saint Vincent and the Grenadines	102039
Samoa	241344
San Marino	33241
Sao Tome and Principe	200063
Saudi Arabia	38799405
Senegal	19697326
Serbia	6857047
Seychelles	110425
Sierra Leone	9594925
Singapore	6151006
Sint Maarten	44842
Slovakia	6721845
Slovenia	2121965
Solomon Islands	812278
Somalia	20519452
South Africa	62452046
South Korea	51738133
South Sudan	11773962
Spain	47916391
Sri Lanka	22128473
Sudan	53475114
Suriname	643799
Sweden	10972682
Switzerland	8987962
Syria	26609578
Taiwan	24061117
Tajikistan	11031842
Tanzania	75929841
Thailand	72164274
Timor-Leste	1447806
Togo	9943306
Tokelau	1925
Tonga	111146
Trinidad and Tobago	1558025
Tunisia	12832841
Turkey	88238740
Turkmenistan	6893706
Turks and Caicos Islands	48776
Tuvalu	11866
Uganda	54881661
Ukraine	25048615
United Arab Emirates	9829739
United Kingdom	68664419
United States	344766399
United States Virgin Islands	96371
Uruguay	3405714
Uzbekistan	37487941
Vanuatu	367517
Vatican City	505
Venezuela	28814808
Vietnam	101873936
Wallis and Futuna	11303
Western Sahara	630963
Yemen	37514851
Zambia	22981544
Zimbabwe	18054551

234 rows in set (0.00 sec)

Q16 Largest Population Country

```
mysql> SELECT
-> `Country/Territory`,
-> `2022 Population`
-> FROM world_population
-> ORDER BY `2022 Population` DESC
-> LIMIT 1;
+-----+-----+
| Country/Territory | 2022 Population |
+-----+-----+
| China             | 1425887337      |
+-----+-----+
1 row in set (0.00 sec)
```

Purpose

Finds the most populated country.

Concepts

- **ORDER BY DESC**
- **LIMIT 1**

Q17. Smallest Population Country

```
mysql> SELECT
-> `Country/Territory`,
-> `2022 Population`
-> FROM world_population
-> ORDER BY `2022 Population` ASC
-> LIMIT 1;
+-----+-----+
| Country/Territory | 2022 Population |
+-----+-----+
| Vatican City      | 510             |
+-----+-----+
1 row in set (0.00 sec)
```

Purpose

Finds the least populated country.

Concepts

- **ORDER BY ASC**
- **LIMIT**

Q18 Highest Density Country

Purpose

Finds the country with highest population density.

```
mysql> SELECT
-> 'Country/Territory',
-> 'Density (per km²)',
-> FROM world_population
-> ORDER BY 'Density (per km²)' DESC
-> LIMIT 1;
+-----+-----+
| Country/Territory | Density (per km²) |
+-----+-----+
| Macau             | 23172.2667        |
+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

Q19 Continent Wise Total Population

```
mysql> SELECT
-> Continent,
-> SUM('2022 Population') AS total_population
-> FROM world_population
-> GROUP BY Continent;
+-----+-----+
| Continent | total_population |
+-----+-----+
| Asia      | 4721383274       |
| Europe    | 743147538        |
| Africa    | 1426730932       |
| Oceania   | 45038554         |
| North America | 600296136       |
| South America | 436816608       |
+-----+-----+
6 rows in set (0.00 sec)
```

Purpose

Calculates total population of each continent.

Functions Used

SUM()

GROUP BY